

THE P-NORM

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ABSTRACT. This is a small article dedicated to showing that the p -norm on $\mathbb{R}^n$ is a norm. In particular, we will be using an argument involving convexity to show Minkowski's inequality. I am indebted to Michelle Daher for expositing this in MA260 lectures at the University of Warwick.

1. PRELIMINARIES

Before investigating the p -norm, we will recall a few definitions for completeness.

Definition. Let X be a vector space over $\mathbb{R}$. A *norm* is a function

$$\|\cdot\| : X \times X \rightarrow \mathbb{R}^+$$

Such that the following three conditions hold.

- (1) $\|x\| = 0$ if and only if $x = 0$
- (2) $\|\lambda x\| = |\lambda| \|x\|$ for all $\lambda \in \mathbb{R}$, $x \in X$.
- (3) $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in X$. (The Triangle Inequality)

A vector space with such a norm function is called a *normed vector space*, and is denoted by the tuple $(X, \|\cdot\|)$.

Definition. Let $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. The p -norm, on $\mathbb{R}^n$ is given by the following.

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}, \quad p \in [1, \infty)$$

It is clear that the p -norm satisfies properties (1) and (2) of being a metric. The triangle inequality (often called *Minkowski's Inequality*) in this case is much less clear. It is certainly unclear how to "brute-force" this, as we often do with the euclidean norm. Thankfully, we can reinterpret the triangle inequality to have a nice geometric interpretation, which makes it easier to work with.

Definition. Let $(X, \|\cdot\|)$ be a normed vector space. The *unit ball* is the following set.

$$\mathcal{B} = \{x \in X \mid \|x\| \leq 1\}$$

Definition. Let X be a set. We say that X is *convex* if for all $x, y \in X$ we have that

$$\lambda x + (1 - \lambda)y \in X \quad \text{for all } \lambda \in [0, 1]$$

What the above definition is trying to capture is that a set is convex whenever it contains the line segment between two points. For instance, the unit ball with respect to the euclidean norm in $\mathbb{R}^2$ is convex, whereas $\{0\} \cup \{1\} \subset \mathbb{R}$ is not, as it doesn't contain 0.5, for instance.

2. THE GEOMETRY OF NORMS

We now get onto the first interesting proposition.

Proposition. *Let X be a vector space, $\|\cdot\| : X \times X \rightarrow \mathbb{R}^+$ a function that satisfies properties (1) and (2) of being a norm. Then $\|\cdot\|$ satisfies the triangle inequality if and only if $\mathcal{B}$ is convex.*

Proof. Suppose firstly that $\|\cdot\|$ satisfies the triangle inequality. Let $x, y \in \mathcal{B}$. Then we have the following:

$$\begin{aligned} \|\lambda x + (1 - \lambda)y\| &\leq |\lambda| \|x\| + |1 - \lambda| \|y\| && \text{by the triangle inequality.} \\ &= \lambda \|x\| + (1 - \lambda) \|y\| && \text{as } \lambda \in [0, 1] \\ &\leq \lambda + 1 - \lambda \\ &= 1 \end{aligned}$$

Therefore, the line segment is contained inside the unit ball.

Now suppose that the unit ball is convex. Let $x, y \in X$. If one of x or y is zero then the triangle inequality trivially holds, so we can suppose that both are nonzero. Now, as both are non-zero, we have the following:

$$\frac{x}{\|x\|}, \frac{y}{\|y\|} \in \mathcal{B}$$

And taking

$$\lambda = \frac{\|x\|}{\|x + y\|} < 1, \quad 1 - \lambda = \frac{\|y\|}{\|x + y\|}$$

We get the following:

$$\begin{aligned} &\left\| \frac{x}{\|x\|} \frac{\|x\|}{\|x + y\|} + \frac{y}{\|y\|} \frac{\|y\|}{\|x + y\|} \right\| \leq 1 \\ \implies &\frac{\|x + y\|}{\|x\| + \|y\|} \leq 1 \\ \implies &\|x + y\| \leq \|x\| + \|y\| \end{aligned}$$

□

I really like the above proposition. I enjoy that it links geometry and analysis, and it teaches a lesson on how we can be better at doing analysis - often drawing the right picture will lead us to the correct argument.

Definition. We say that $f : [a, b] \rightarrow \mathbb{R}$ is *convex* if for all $x, y \in [a, b]$ we have

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

We say $f : \mathbb{R} \rightarrow \mathbb{R}$ is *convex* if it is convex on all intervals $[a, b]$.

In other words, what we are saying is that the graph of the function lies below the line segment that connects points on said graph, or that the set of points "above" the graph is convex. The above definition is not easy to work with, so we have the following proposition.

Proposition. *Suppose that f is a real valued function such that $f \in C^2([a, b])$. Assume further that $f'' \geq 0$ on $[a, b]$. Then f is convex on $[a, b]$.*

Proof. By rescaling, we can assume this interval to be $[0, 1]$. More fully, we want to show that for all $c \in [0, 1]$,

$$f(c) \leq cf(0) + (1 - c)f(1) \text{ for all } \lambda \in [0, 1]$$

By the mean value theorem, there exists $a \in [0, c]$, $b \in [c, 1]$ such that

$$f'(a) = \frac{f(c) - f(0)}{c}$$

$$f'(b) = \frac{f(1) - f(c)}{1 - c}$$

As $f'' \geq 0$, we have that $f'(a) - f'(b) \geq 0$. Moreover, as $0 \leq c \leq 1$, we have that $c(1 - c) \geq 0$. Therefore,

$$cf(1) - f(c) + (1 - c)f(0) = c(1 - c)(f'(b) - f'(a)) \geq 0$$

And this implies the result. $\square$

Corollary. For $p \in [1, \infty)$, the function $t \mapsto |t|^p$ is convex on $\mathbb{R}$

Proof. If $p = 1$, this is clear. Otherwise suppose not. Then there exists an interval on which it is not convex, say $[x, y]$. However, on this interval, we have that $f'' \geq 0$, which contradicts the above proposition. $\square$

We are now ready to bring everything together and prove the main result.

Theorem. (Minkowski's Inequality) Let $\|\cdot\|_p$ denote the p -norm on $\mathbb{R}^n$. Then for all $x, y \in \mathbb{R}^n$, we have

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p$$

Proof. We will show that the unit ball is convex. Let $x, y \in \mathcal{B}$. We want to show that the line segment is also contained in the unit ball. This is a straightforward verification using the above corollary. For all $\lambda \in [0, 1]$, we have

$$\begin{aligned} \|\lambda x + (1 - \lambda)y\|_p^p &= \sum_{i=1}^n |\lambda x_i + (1 - \lambda)y_i|^p \\ &\leq \sum_{i=1}^n \lambda |x_i|^p + (1 - \lambda) |y_i|^p \\ &= \lambda \|x\|_p^p + (1 - \lambda) \|y\|_p^p \\ &\leq \lambda + (1 - \lambda) = 1 \end{aligned}$$

$\square$

Corollary. The p -norm is a norm on $\mathbb{R}^n$